Exploratory Data Analysis (EDA)

The goal of this work was to attempt to understand electric vehicle charging duration and find any impactful variables in the dataset.

The outcome of the EDA on the ev_battery_charging_data set did not find any initial, strong relationships. A variety of comparisons were done, including:

- PMF to compare charging cycles between two EV models
- CDF to compare charging duration between two EV models
- Analytical distribution on charging duration
- Scatter plot comparing charging duration vs charging cycles
- Scatter plot comparing charging duration vs state of charge
- Pearson correlation of the two scatter plots
- Hypothesis test comparting mean charging duration between fast and slow charging
- Regression analysis between charging duration and current amperage

Lessons Learned

This dataset did not help us answer our question; however, we got a better understanding of the data and what to consider for next steps. I feel that a key missing variable is battery capacity. There are many other impacts to battery life (such as technology, driver usage, region) that could be prioritized and explored.

There was one normal distribution found in Degradation Rate out of the six total variables. I am curious to see the relationship of this dataset variables and battery capacity. There should be more insightful information discovered if we are able to include this data.

I was expecting to find more initial relationships between the variables. I was shocked to not find many differences when doing comparisons with temperature specifically. More questions arose around further analysis:

- What relationships are strong with State of Charge variable?
- What impacts do ambient temperature have on battery capacity and charge duration?
- What additional information can we get regarding charging cycle and what impacts it's diverse frequency counts?

As for challenges, I was having trouble working with the book author's libraries to accomplish the assignment. Therefore, I had to research how to implement in Python. I had high hopes in working with this dataset, but I am now interested in further researching how to better understand EV battery life over time.